

PRACTICAL WEEK7

TASK 1: Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
GNU nano 8.7.1 memory_layout.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void display() {
    printf("Inside code segment\n");
}

int main() {
    int stack_var = 30;
    static int local_static = 40;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n", getpid());

    printf("Code address      : %p\n", (void *)display);
    printf("Global variable    : %p\n", (void *)&global_var);
    printf("Static variable       : %p\n", (void *)&static_var);
    printf("Local static variable : %p\n", (void *)&local_static);
    printf("Heap variable         : %p\n", (void *)heap_var);
    printf("Stack variable        : %p\n", (void *)&stack_var);

    printf("\nProcess is running... Press Ctrl+C to stop.\n");

    while (1) {
        sleep(2);
    }

    free(heap_var);
    return 0;
}
```

```
rishill10@ASUS16X:~$ mkdir Week7 Practical
rishill10@ASUS16X:~$ cd Week7 Practical
-bash: cd: too many arguments
rishill10@ASUS16X:~$ rmdir Week7 Practical
rishill10@ASUS16X:~$ mkdir Week7_Practical
rishill10@ASUS16X:~$ cd Week7_Practical
rishill10@ASUS16X:~/Week7_Practical$ pwd
/home/rishill10/Week7_Practical
rishill10@ASUS16X:~/Week7_Practical$ nano memory_layout.c
rishill10@ASUS16X:~/Week7_Practical$ gcc memory_layout.c -o memory_layout
./memory_layout
Process ID: 1743
Code address      : 0x580efd3791c9
Global variable    : 0x580efd37c010
Static variable    : 0x580efd37c014
Local static variable : 0x580efd37c018
Heap variable      : 0x580f12c63010
Stack variable     : 0x7ffd3d96150c

Process is running... Press Ctrl+C to stop.
^C
rishill10@ASUS16X:~/Week7_Practical$ |
```

TASK 2: Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
rishill10@ASUS16X: ~/Week7_ × + v
GNU nano 8.7.1 memory_analysis.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void display() {
    printf("Inside code segment\n");
}

int main() {
    int stack_var = 30;
    static int local_static = 40;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n", getpid());

    printf("Code address      : %p\n", (void *)display);
    printf("Global variable     : %p\n", (void *)&global_var);
    printf("Static variable        : %p\n", (void *)&static_var);
    printf("Local static variable  : %p\n", (void *)&local_static);
    printf("Heap variable          : %p\n", (void *)heap_var);
    printf("Stack variable         : %p\n", (void *)&stack_var);

    printf("\nProcess is running... Press Ctrl+C to stop.\n");

    while (1) {
        sleep(2);
    }

    free(heap_var);
    return 0;
}
```

```
rishill10@ASUS16X: ~/Week7_ × + ▾
rishill10@ASUS16X:~/Week7_Practical$ nano memory_analysis.c
rishill10@ASUS16X:~/Week7_Practical$ gcc memory_analysis.c -o memory_analysis
rishill10@ASUS16X:~/Week7_Practical$ ./memory_analysis &
[1] 1798
Process ID: 1798
Code address      : 0x5a194432e1c9
Global variable   : 0x5a1944331010
Static variable   : 0x5a1944331014
Local static variable : 0x5a1944331018
Heap variable     : 0x5a1959256010
Stack variable    : 0x7ffcc22a77dc

Process is running... Press Ctrl+C to stop.
rishill10@ASUS16X:~/Week7_Practical$ cat /proc/1798/maps
5a194432d000-5a194432e000 r--p 00000000 08:30 42057 /home/rishill10/Week7_Practical/memory_an
alysis
5a194432e000-5a194432f000 r-xp 00001000 08:30 42057 /home/rishill10/Week7_Practical/memory_an
alysis
5a194432f000-5a1944330000 r--p 00002000 08:30 42057 /home/rishill10/Week7_Practical/memory_an
alysis
5a1944330000-5a1944331000 r--p 00002000 08:30 42057 /home/rishill10/Week7_Practical/memory_an
alysis
5a1944331000-5a1944332000 rw-p 00003000 08:30 42057 /home/rishill10/Week7_Practical/memory_an
alysis
5a1959256000-5a1959277000 rw-p 00000000 00:00 0 [heap]
775000c00000-775000c28000 r--p 00000000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
775000c28000-775000dc0000 r-xp 00028000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
775000dc0000-775000e0e000 r--p 001c0000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
775000e0e000-775000e12000 r--p 0020d000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
775000e12000-775000e14000 rw-p 00211000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
775000e14000-775000e21000 rw-p 00000000 00:00 0
775000f05000-775000f08000 rw-p 00000000 00:00 0
775000f0e000-775000f10000 rw-p 00000000 00:00 0
775000f10000-775000f14000 r--p 00000000 00:00 0
775000f14000-775000f16000 r--p 00000000 00:00 0
775000f16000-775000f18000 r-xp 00000000 00:00 0 [vvar]
775000f18000-775000f19000 r--p 00000000 08:30 13931 [vvar_vclock]
.so.2 [vdso]
775000f19000-775000f48000 r-xp 00001000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
775000f48000-775000f53000 r--p 00030000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
775000f53000-775000f55000 r--p 0003b000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
775000f55000-775000f56000 rw-p 0003d000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
775000f56000-775000f57000 rw-p 00000000 00:00 0
7ffcc2289000-7ffcc22aa000 rw-p 00000000 00:00 0 [stack]
rishill10@ASUS16X:~/Week7_Practical$ |
```